TAO Wenwen

Curriculum Vitae — PhD and Graduate Admissions

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Research Profile

Clinically trained multimodal human-neuroscience researcher studying youth mental health and developmental variation in brain–behavior relationships. My trajectory has progressed from clinical cohort implementation and basic MRI/EEG quality control, through hands-on neuroimaging analysis, to independent design of synchronized EEG–fNIRS, physiological, behavioral and clinical research systems.

I am particularly interested in distinguishing stable vulnerability from within-person state change and using longitudinal, clinically grounded multimodal data to inform individualized assessment and intervention.

Research Interests

Youth mental health and developmental psychopathology · Multimodal brain–behavior dynamics · Within-person and longitudinal variation · Emotion regulation, cognitive control, risk and resilience

Education

B.M. in Clinical Medicine | Nanchang University **2017–2022**

GPA 3.2/4.0 (82/100) · Psychiatry, neurology, medical statistics and evidence-based medicine

Research Experience

Research Assistant | Tsinghua University, CogDrive Lab **Mar 2026–Present**

State Key Laboratory of Intelligent Green Vehicle and Mobility · Advisor: Prof. Dongpu Cao · Clinical collaboration: Tongji Hospital, HUST

Multimodal developmental neuroscience of youth depression and translation to intelligent health-cockpit research

- Investigate how neural, hemodynamic, autonomic, behavioral and clinical signals can characterize mental and emotional states across ages 6–18 while accounting for developmental differences.
- Independently designed the study protocol, cognitive-affective paradigms and clinician-led interview workflow for a planned 300-participant clinical cohort and 150 healthy controls; the protocol received joint scientific review and hospital IRB approval.
- Designed the synchronized acquisition and event-trigger architecture integrating simultaneous 64-channel EEG–fNIRS, eye tracking, facial video, speech, ECG, PPG, GSR and respiration during computerized tasks and clinical interviews, with acquisition planned around downstream preprocessing, neurovascular-coupling and multimodal modeling requirements.
- Established age-adaptive clinical ground truth using CDRS-R, self/parent reports, K-SADS-PL DSM-5 confirmation and multisource severity grading, including developmentally appropriate consent, assent and risk-escalation procedures.
- Built BPlus MVP-0 to operationalize Rest, MMN, VFT, N-back, Emotion and Recovery modules with response/event logging, preflight checks and QC/state reports. Software demonstrations and dry runs passed; real-hardware synchronization and formal pilot data collection remain pending.
- Designed the health-cockpit translation pathway to connect validated laboratory markers with lower-burden in-vehicle sensing, confidence-aware outputs and safety-gated human–AI interaction; clinical validation and vehicle deployment remain future stages.

Research Intern | Tsinghua University, Yan Lab **Sep 2025–Mar 2026**

Advisor: Prof. Chaogan Yan · Mind Flower depression research

Depression neuroimaging, repeated symptom measurement and individualized rTMS

- Examined individual differences in brain function, cognition and repeated symptom trajectories relevant to understanding depression and treatment response, combining clinical phenotyping, MRI, cognitive tasks, rTMS records and wearable follow-up.
- Conducted MINI and serial HAMD/HAMA/MADRS assessments and managed resting-state, rumination and distraction MRI sessions, including individualized cue preparation, E-Prime task adaptation, head-motion monitoring (<3 mm) and OptiMRI audio QC.
- Completed structured training in DPABI/DPABISurf neuroimaging workflows and developed greater hands-on depth in resting-state and task-fMRI preprocessing, quality control and participant-level interpretation beyond the basic QC work of earlier projects.
- Linked repeated HAMD-6 and rTMS tolerability records with WCST, SST, emotional n-back and two-week HRV, sleep and activity monitoring; standardized E-Prime recovery, export and hierarchical NAS archiving across repeated visits.

Visiting Research Intern | Beijing Normal University, Liu Lab **Sep 2025–Mar 2026**

Advisor: Dr. Zheyun Liu

MRI-guided transcranial ultrasound stimulation of bilateral GPi in Parkinson's disease

- Supported individualized bilateral GPi targeting using structural MRI and Brainsight neuronavigation within a multidisciplinary TUS workflow.
- Operated an approximately 1.5-hour TUS–cognitive assessment protocol focused on memory and cognitive flexibility, documenting pulse parameters, behavioral responses, safety and participant-reported auditory/thermal tolerability.

Research Assistant | Shenzhen Mental Health Center, Addiction Medicine **Jan 2024–Aug 2025**

Advisors: Prof. Mei Yang, Prof. Xiaojian Jia, Dr. Gengdi Huang and Dr. Yi Zhang

Addiction neuroimaging and longitudinal clinical cohorts

- Investigated how clinical, circuit-level and biological factors relate to addiction severity and longitudinal outcomes in alcohol and methamphetamine use disorders.
- Coordinated recruitment, screening, psychiatric assessment and longitudinal follow-up for 290+ participants, working across clinicians, department leadership, radiology staff and technical engineers to support MRI/DTI, EEG, behavioral and biospecimen workflows; maintained >95% data completeness.
- Participated in MRI/DTI and EEG acquisition, quality control and basic preprocessing, and built multimodal participant, sample, metadata and anomaly-tracking procedures that supported reproducible downstream analysis.
- Integrated 16S rRNA, GC–MS SCFAs and MRI in a 220-participant AUD project; identified three candidate microbial markers associated with SCFA levels and observed correlations between lower SCFAs and lower FA in the anterior limb of the internal capsule and corpus callosum.
- Coordinated a 160-participant TMP imaging study and assessed pre/post FA in the corpus callosum and superior longitudinal fasciculus; preliminary differences require longitudinal confirmation, and the project produced an optimized DTI + resting-state fMRI acquisition and QC workflow.
- Independently acquired MRI/EEG from 10 participants with methamphetamine dependence, maintained an 80-case database and contributed to DTI/fMRI analyses of prefrontal–hippocampal, cingulum and fornix pathways.
- Examined PFC–ventral striatal structure–function relationships in 20 deeply phenotyped cases and developed an exploratory logistic-regression relapse model, generating hypotheses for future connectivity-informed neuromodulation studies.

Publications and Manuscripts

- Tao, W. Early Depressive Symptom Severity and Six-Month Return to Drinking After Inpatient Treatment for Alcohol Dependence: A Prospective Cohort Study. Manuscript submitted to Psychiatry Research. First author (not co-first author); prospective cohort and longitudinal outcome analysis.
- Tao, W. Temporal Dataset Shift Limits Relapse-Risk Stratification Across the Inpatient-to-Community Transition in Alcohol Dependence: A Temporal Comparison of Conventional Machine Learning and Self-Supervised Representation Learning. Submitted to Addiction Science & Clinical Practice (ASCP). First author (not co-first author); temporal validation and comparative modeling.
- Tao, W. Task state dominates participant-held-out prediction of risky decisions in methamphetamine dependence, with limited incremental value from pre-decision EEG. Submitted to Addictive Behaviors. First author (not co-first author); participant-held-out EEG and behavioral prediction analysis.

Clinical Experience

Resident Physician | Wannian County Hospital of TCM **Dec 2022–Dec 2023**

- Managed 15–20 chronic-disease inpatients through assessment, treatment planning and follow-up, developing first-hand understanding of heterogeneity in real-world clinical presentations and the need for clinically interpretable research measures.

Clinical Intern | Ganzhou People's Hospital **Feb 2021–Feb 2022**

- Completed 12 rotations, including Neurology ICU and Cardiology ICU; performed supervised lumbar puncture and ECG interpretation and participated in routine inpatient and perioperative workflows.

Additional Technical Experience

Machine Vision Engineering Intern | OPT **Nov 2025–Feb 2026**

- Worked with OpenCV, PLC, STM32 and FPGA-based machine-vision systems, developing experience in real-time data acquisition, hardware–software integration and signal-trigger logic.
- Applied this engineering perspective to later multimodal synchronization, event-trigger design and experimental-operations software, while keeping machine vision secondary to the clinical-neuroscience research trajectory.

Methods and Technical Expertise

Multimodal experimental design: Protocol and paradigm development; TTL/LSL and event-trigger architecture; synchronized acquisition; clinical interview workflow; multimodal QC; analysis-ready data specification

Neuroimaging analysis: Structural MRI; resting-state and task fMRI; DTI tractography; FA/ADC; structural–functional connectivity; DPABI/DPABISurf; FSL; MRI-based targeting

Neurophysiology and physiology: EEG (32/64-channel); simultaneous EEG–fNIRS; ECG; PPG; GSR; respiration; eye tracking; signal QC; exploratory neurovascular coupling

Clinical and longitudinal research: MINI; K-SADS-PL DSM-5; CDRS-R; HAMD/HAMA/MADRS; recruitment; repeated assessment; participant follow-up; cohort and IRB coordination; pediatric safeguards

Quantitative and computational methods: Python; MATLAB; R; SPSS; regression/classification; cross-validation; AUC; participant-held-out and temporal validation; longitudinal/repeated-measures analysis

Translational systems: BPlus MVP-0; response/event logging; preflight and QC reporting; longitudinal databases; Data Dictionary development; multicenter harmonization; health-cockpit research architecture

Additional modalities: rTMS; TUS; Brainsight neuronavigation; 16S rRNA microbiome sequencing; GC–MS SCFA metabolomics; biospecimen SOPs

English — TOEFL 97; IELTS 6.5 | Chinese — Native